CellLab

Touch-free 3D cell explorer, controlled by your hands

Stack

Three.js + MediaPipe Hands + vanilla JavaScript

A web app where you rotate, zoom and select the parts of a 3D animal cell using only your webcam and hand gestures. No controller, no install, one HTML file.

TKS Focus Project | Build phase | Author: Ulkhad

The idea

Learn cell biology by touching it, without touching anything

- Problem: cell diagrams in textbooks are flat and passive. Learners rarely get to explore parts from every angle.
- Solution: a real-time 3D cell that responds to your real hands, tracked through an ordinary webcam.
- Result: three connected pages in one dashboard.

1. Explore

Interactive 3D cell. Point, pinch, rotate and zoom with hand gestures (or the mouse).

2. Cell guide

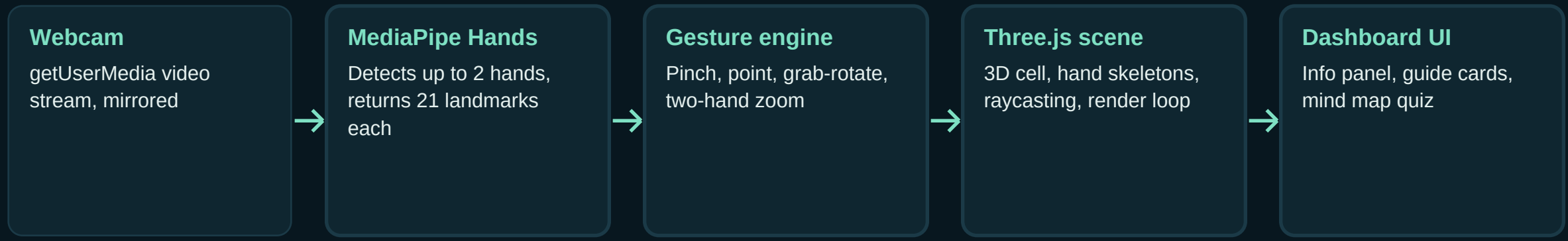
Definition, function and a memory tip for each of the 8 organelles, with search.

3. Mind map test

Bubbles describe a job. Pick which organelle does it. Score and mistakes are tracked.

Architecture

Everything runs in the browser, nothing is sent to a server



- Data flow: frame -> landmarks -> gesture state -> scene update -> pixels, every animation frame.
- Privacy: video is processed locally. The model files are downloaded once from a CDN.

Technology stack

Technology	Role	Source
Three.js r128	3D rendering: meshes, lights, raycaster, scene graph	cdnjs
MediaPipe Hands 0.4	Hand landmark detection (WASM + ML model), 21 points per hand	jsDelivr
WebRTC getUserMedia	Webcam access in the browser	built in
HTML + CSS + vanilla JS	UI, navigation, guide, quiz. No framework, no build step	built in
Blender (upstream)	Rigging a 21-bone hand mesh for the next version	local tool

Why this stack: one file, zero setup, runs on any laptop with a webcam. Easy to host on GitHub Pages.

Code structure

cell-lab.html is organised in labelled sections

`O  (data)`	8 organelles: name, color, definition, function, tip, quiz clue. One source feeds all 3 pages.
`3D scene`	Builds the cell from spheres and tori, one Group per organelle with a key.
`Picking`	pick(x,y): raycast from the camera. Small organelles win over nucleus and membrane.
`Mouse fallback`	Drag, wheel and click so the app works without a camera.
`Hand tracking`	onRes(): draws skeletons, detects pinch, drives rotation, zoom, selection.
`Render loop`	Idle spin, smooth zoom, hover and select glow, then render.
`Guide + Quiz`	cards() builds the guide. build(), ask(), ans() run the mind map test.

Gesture logic

How raw landmarks become interactions

- Mirror: x = 1 - landmark.x so the screen behaves like a mirror.
- Pinch ratio = dist(thumb tip 4, index tip 8) / dist(wrist 0, middle MCP 9). Dividing by hand size makes it work at any distance.
- Hysteresis: pinch starts below 0.35 and ends above 0.55, so it does not flicker.
- Point: raycast from the index fingertip. Pinch on a part selects it. Pinch in empty space grabs.
- Grab: palm movement (landmark 9) rotates the cell. Two grabbing hands: change in distance zooms (0.5x to 2.2x).

```
const sz = dist(lm[0], lm[9]);
const pr = dist(lm[4], lm[8]) / sz;
s.p = was ? pr < .55 : pr < .35;

const k = pick(ndcX(lm[8]), ndcY(lm[8]));
if (s.p && !was) {
  if (k) select(k);   // pinch on part
  else   s.g = 1;     // grab space
}
if (s.p && s.g) {
  G.rotation.y += dx * 7;
  G.rotation.x += dy * 7;
}
```

MediaPipe 21 landmarks

The same indices will drive the Blender rig

Wrist	0
Thumb	1 CMC, 2 MCP, 3 IP, 4 TIP
Index	5 MCP, 6 PIP, 7 DIP, 8 TIP
Middle	9 MCP, 10 PIP, 11 DIP, 12 TIP
Ring	13 MCP, 14 PIP, 15 DIP, 16 TIP
Pinky	17 MCP, 18 PIP, 19 DIP, 20 TIP

- Each hand = 21 points, 20 bones. Bones are named by the landmark at their head (00_WRIST ... 20_PINKY_TIP).
- Webcam "Left" is often the user's right hand when the image is mirrored, so left and right are swapped in code.
- Landmark z is relative to the wrist and rough. It is scaled by 6 for a depth hint.

Run it and what comes next

Run locally

1. Open cell-lab.html in Chrome or Edge. 2. Click Start hand tracking, allow the camera. 3. If blocked: python -m http.server, then open localhost:8000/cell-lab.html.

Known limits

Hands are skeletons, not the Blender mesh yet. Quiz is mouse-only. English only. Needs internet for the CDN libraries and model. Lighting affects tracking quality.

Roadmap

- Load the rigged GLB hands (GLTFLoader) and drive bones 00-20 from landmarks.
- Hand control on the guide and quiz pages.
- Bundle libraries and model for offline use.
- French / English toggle.
- Plant cell, organelle zoom and labels.
- Smoothing filter on landmarks.
- WebXR and mobile support.